

Explainable Customer Churn Prediction using Apache Spark & XAI

Executive Summary & Technical Architecture Report

1. Executive Highlights & Financial Exposure

Total Ingested Customers: 7,043 | Baseline Churn Rate: 26.54%
Monthly Recurring Revenue at Risk: \$139,130.85 (\$1,669,570.20 annualized), representing 30.5% of revenue.
Champion Machine Learning Model: Gradient Boosted Trees with ROC-AUC of 0.8464.

2. Machine Learning Benchmark Performance

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	80.6%	0.668	0.537	0.596	0.8457
Random Forest	80.1%	0.672	0.487	0.564	0.8436
Gradient Boosted Trees	80.4%	0.668	0.521	0.586	0.8464

3. Explainable AI (SHAP) Insights

- Month-to-Month Contract Status is the single highest contributor towards customer departure.
- High Tenure acts as a strong protective anchor against attrition.
- High Monthly Charges without bundled Tech Support exacerbate risk.
- Electronic Check users present high churn due to manual payment friction.

4. Executive Strategic Recommendations

[P1 - Critical] Month-to-Month Contract Transition

Action: Incentivize 1-year commitments with a 15% discount or bundled streaming benefits; implement automated renewal nudges at month 3 and month 6.

[P1 - Critical] Fiber Optic Value & Support Gap

Action: Bundle complimentary TechSupport and Online Security into high-speed fiber tiers to protect high-ARPU subscribers.

[P2 - High] Electronic Check Payment Friction

Action: Offer a one-time \$10 credit to migrate electronic check customers to automated credit card / ACH autopay.

[P3 - Medium] New Customer Lifecycle Onboarding

Action: Deploy a high-touch 90-day onboarding program, automated satisfaction check-ins, and priority support routing for new signups.